

Project Design Phase

Solution Architecture

Date	14 JUNE 2026
Team ID	INDIVIDUAL PROJECT
Project Name	ShopEZ: E-commerce Application
Maximum Marks	4 Marks

Solution Architecture:

Architectural Blueprint

- **Stack:** Built on a decoupled MERN stack (MongoDB, Express.js, React, Node.js) separating UI from backend logic.
- **Frontend:** Mobile-first React Single Page Application (SPA) designed to load instantly on smartphones.
- **Backend:** Node.js and Express.js runtime handling asynchronous, non-blocking requests so the app never freezes.
- **Database:** Cloud-hosted MongoDB cluster using Mongoose schemas for flexible, document-based data management.

Technical Performance

- **Query Indexing:** Applies single and compound database indexing to eliminate search filter lag.
- **Target Latency:** Engineered to process and return API responses in under 200ms.
- **Stateless Security:** Employs secure JSON Web Tokens (JWT) in API headers to enable a fast, safe, single-step checkout.

Development Roadmap

- **Phase 1:** Database schema design, Mongoose modeling, and core Express API configuration.
- **Phase 2:** Integration of JWT route guards, one-step checkout validation, and dynamic order tracking.
- **Phase 3:** Mobile viewport optimization, compound index fine-tuning, and heavy session stress testing.

